PRD: Cargo Schedule Demo (Phasis Air Cargo)

Oct 17, 2026 · @Mikheil Tenieshvili

1. Goals, non-goals and pitch success criteria

The demo exists to win a second meeting with a Georgian cargo airline: in under 10 minutes the prospect watches a messy Excel schedule become a validated schedule, a cleared conflict list and a flight packet. It is a frontend-only Next.js app deployed on Vercel, with all data seeded in the browser. The product name in the design is AeroPlanner and the fictional operator is Phasis Air Cargo (ICAO PHC, IATA P7, base UGTB/TBS).

Goals

- G1. Show that messy Excel schedule strings parse into structured legs, and that dirty cells are surfaced for review instead of silently guessed.
- G2. Show that the system catches operational mistakes (duplicate flight number, aircraft teleportation, aircraft in AOG, permit gaps, overlaps) before the aircraft does, each with a one-click fix.
- G3. Show a consolidated flight packet (PDF, Excel, Word) built from picked legs.
- G4. Feel like a product the airline could run next Monday: realistic data, a fast grid, no dead buttons inside the three storylines.
- G5. Be runnable by anyone on the team from a URL, repeatable, with deterministic data.

Non-goals

- NG1. No real authentication. One fixed user, "Ops planner", and no login screen.
- NG2. No integration with permit authorities, email, maintenance, crew or flight-planning systems. Permit numbers are fake.
- NG3. No real database and no API. State is seeded in the browser and edits live in the tab session only.
- NG4. Not an operational tool. The checklist has no regulatory validity, and there is no weight and balance, performance, NOTAM or weather.
- NG5. No multi-user features: no comments, notifications or audit log.
- NG6. English only, desktop first (1280 px wide and up). No phone layout.

- NG7. No generic Excel column-mapping wizard. The parser targets the one sheet layout in section 7.

Success criteria for the pitch

ID	Criterion	How it is checked
SC1	After Reset to full demo the app shows 88 active legs, 2 cancelled legs, 9 open issues (7 errors, 2 warnings) and 3 urgent permit rows.	Automated test, plus a manual check before every pitch
SC2	Storyline A: 90 legs parsed from 118 rows, 14 flagged, 76 ready. All 14 are resolved in the UI and 90 legs are imported.	Automated test, presenter time 4 min or less
SC3	Storyline B: open issues fall from 9 to 6 after fixing R1, R2 and the R3 AOG issue. The three fixes create no new issue.	Automated test, presenter time 4 min or less
SC4	Storyline C: three selected legs export a consolidated PDF in 3 s or less.	Stopwatch on a mid-range laptop, presenter time 2 min or less
SC5	Zero console errors and zero dead-end clicks in any storyline. Reset returns the app to the known state in 1 s or less.	Scripted run of all three storylines
SC6	The Schedule grid with 1,000 legs scrolls at 55 fps or better and a full validation run takes 100 ms or less.	Performance test in section 10
SC7	Pitch outcome (target, review after five pitches): at least 3 of 5 prospects agree to send a redacted real schedule for a parser trial within 14 days.	Sales log

The demo clock is fixed at Sun 13 Sep 2026, 14:00Z, so every 48-hour rule gives the same result on every run.

2. Personas and demo storylines

The demo speaks to the people who build and defend the schedule every day, and to the manager who pays for the mistakes. Each storyline starts from a named reset state, so none depends on another having run.

Persona	Job in the airline	Pain today	Where the demo meets them
Ops planner (the logged-in user, "OP Ops planner")	Builds the weekly and seasonal schedule in Excel	Typos and stale cells only show up when an aircraft is already late or in the wrong place	Import, Schedule, Conflicts centre
Permits coordinator	Requests landing and overflight permits, chases pending ones	No single list of what is missing for flights in the next 48 hours	Permits board, permit rows in the leg drawer
Head of operations (decision maker in the room)	Signs off the schedule, answers for cancellations	Cannot see risk without asking three people	Overview, Conflicts centre
Maintenance controller	Owns AOG and C-Check windows	Planners learn about a grounded aircraft late	Fleet availability strip, Gantt hatching
Commander and crew	Fly the legs	Receive packets with missing or inconsistent aircraft data	Checklist export
Presenter (our team)	Drives the demo	A demo that breaks or differs between runs	Reset, deterministic seed

Storyline A: From Excel chaos to clean schedule

Start state: Reset to empty schedule (fleet, airports and company profile stay, the schedule is empty). Target time 4 min.

1. Open Import from Excel and choose "Use sample: Phasis_2026_Season.xlsx".

2. Watch the progress bar. The summary reads: 90 legs parsed from 118 rows, 14 need review, 76 ready to import.
3. In the split view (messy cell text on the left, structured legs on the right) point at one clean row, then at a flagged row such as PHC601 on Mon 14 Sep ("DAY 3 LT" where the derived day is 2).
4. Click Accept fix on the 9 flagged rows that carry a suggestion. Open the leg drawer for the 5 rows without one and correct them.
5. When "Need review" reads 0, click Import 90 legs.

End state: the Schedule shows 90 legs, 88 active and 2 cancelled (PHC517 and PHC518 on Sun 27 Sep, shown with a CNL tag).

Storyline B: Catch the mistake before the aircraft does

Start state: Reset to full demo. Target time 4 min.

1. On Overview, "Needs attention" shows 9 open. Click Open Conflicts centre.
2. R1: PHC509 is on both 4L-PXA and 4L-PXB on Fri 18 Sep. Click the fix "Make the 4L-PXB leg PHC509D". Open issues: 8.
3. R2: 4L-PXB flew PHC511 into LIPK/FRL, but PHC512 on Thu 17 Sep departs LIMC/MXP. Click "Set departure to LIPK/FRL". Open issues: 7.
4. R3: 4L-PXB is AOG from Mon 21 Sep 20:00Z to Tue 22 Sep 10:00Z and PHC503 and PHC504 fall inside it. Click "Reassign 2 legs to 4L-PXA". Open issues: 6.
5. Scroll to "Deliberately not flagged" and explain PHC611 (through flight) and PHC507 with PHC507D (parallel flights). Click Go to flight on the R3 card and show the Gantt: 4L-PXA now carries the legs and the AOG block on 4L-PXB is empty.

End state: 6 open issues remain: the R3 C-Check on 4L-PXA, two R4 urgent permits, two R4 missing permits and one R5 overlap.

Storyline C: From schedule to flight packet

Start state: Reset to full demo (or straight after B, since 4L-PXC is untouched). Target time 2 min.

1. Open Checklist export. Tick the two PHC611 legs (TBS to URC, URC to XIY, Wed 16 Sep) and PHC612 XIY to URC (Thu 17 Sep). That is 3 legs on 4L-PXC.
2. The preview shows one consolidated document: company header, one aircraft block, route map TBS, URC, XIY, URC, a 3-row legs table, 4 permit rows and the signature blocks.

3. Click Export PDF. After the progress overlay the file `PHC_Checklist_2026-09-16_to_2026-09-17_3legs.pdf` downloads. Mention that Excel and Word are one click away.

3. Domain glossary

These definitions are the ones the product uses. Where an industry term has several meanings, the right-hand column says which one the demo implements.

Term	Meaning	In this product
ICAO	International Civil Aviation Organization. Four-letter airport codes (UGTB) and three-letter airline designators (PHC).	Primary key for airports. Prefix of every flight number (PHC611).
IATA	International Air Transport Association. Three-letter airport codes (TBS) and two-character airline codes (P7).	Shown next to ICAO as UGTB/TBS. Never used as a key.
UTC	Coordinated Universal Time, written with a Z suffix in the Excel sheet (03.30Z).	Every stored and displayed time, unless tagged LT.
LT	Local time at the airport concerned.	Used only to tag the arrival day ("DAY n LT"). Offsets are in the Airport directory.
ETD	Estimated time of departure.	The planned departure instant stored on the leg, in UTC. The Excel token ETD/03.30Z.
ETA	Estimated time of arrival.	The planned arrival instant, in UTC. ETA minus ETD is the block time. Shown with (+1) when it falls on the next UTC day.
STD	Scheduled time of departure, the published timetable time.	The demo has no separate scheduled and estimated times. An STD token in Excel is read as ETD.

Term	Meaning	In this product
MTOW	Maximum take-off weight, in kg.	A Fleet registry field printed on the checklist.
AOG	Aircraft on ground: unserviceable and waiting for repair or parts.	A MaintenanceBlock of kind aog. Legs inside it raise R3.
C-Check	Heavy scheduled maintenance check that takes an aircraft out of service for days.	A MaintenanceBlock of kind c_check. Legs inside it raise R3.
CNL	Cancelled.	Soft cancellation: the leg stays visible and in history, and is excluded from counts and validation (R7).
Overflight	Crossing a state's airspace without landing there.	Needs an overflight permit for that country.
Landing permit	Authorisation to land a non-scheduled cargo flight in a state.	Needed for every arrival in a country other than Georgia.
Through-flight	Consecutive legs under one flight number on one aircraft, such as PHC611 TBS to URC to XIY.	Grouped with a double bar on the Schedule. Never raises R1.
Ferry leg	A positioning flight without cargo, for example to move an aircraft to where it is needed.	Leg type ferry. Its destination may be the placeholder XXXX/XXX, which skips continuity (R2) and permit (R4) checks.
ISO weekday	Weekday number from ISO 8601: Monday 1 to Sunday 7.	Shown in the "Days + dates + ISO weekday" display mode. In "DAY n LT", n is the ISO weekday of the local arrival date (see open question 2).
Leg	One take-off to one landing by one aircraft.	The unit of the schedule. One row in the data model.
Block time	Time from off-blocks to on-blocks.	ETA minus ETD, in minutes, shown as HH:MM (R6).
Station	An airport used as a schedule point.	Always shown as ICAO/IATA.

4. Data model and derived fields

Only seven entities are stored. Block time, (+1), "DAY n LT", through-flight groups, required permits and every ValidationIssue are derived from them on each change and are never stored. All times are UTC instants; dates are UTC calendar dates.

```
type IsoInstant = string;   // UTC, e.g. "2026-09-16T03:30:00Z"
type IsoDate = string;      // UTC calendar date, e.g. "2026-09-16"
type CountryCode = "GE" | "CN" | "KZ" | "RS" | "GR" | "HU" | "RO" | "IT" |
"AE" | "IN" | "HK" | "XX";

interface Company {
  id: string;                 // "phc"
  legalName: string;          // "Phasis Air Cargo"
  vatNumber: string;
  icao: string;               // "PHC"
  iata: string;               // "P7"
  aocNumber: string;          // "DEMO-0001"
  baseIcao: string;           // "UGTB"
  address: string;
  opsContact: string;
  checklistHeaderText: string;
  logoDataUrl: string | null;
  minGroundMinutes: number;   // 60, used by the R5 fix
}

type AircraftType = "B737-800SF" | "B737-800BCF" | "B767-300ER";
interface Aircraft {
  id: string;                 // same as registration, e.g. "4L-PXA"
  registration: string;
  type: AircraftType;
  serialNo: string | null;
  yearOfManufacture: number | null;
  mtowKg: number | null;
  cargoPayloadKg: number | null;
  status: "active" | "aog" | "maintenance";   // registry status at the demo
clock
  defaultBackupId: string | null;
}
// A registry row is "Incomplete" when serialNo, yearOfManufacture, mtowKg or
cargoPayloadKg is null.
// Null fields print as "Missing" on checklists.
```

```
interface Airport {
  icao: string;               // key, e.g. "UGTB"
  iata: string;               // "TBS"
  name: string;
  countryCode: CountryCode;
  utcOffsetMinutes: number;   // fixed offset, no daylight saving in the demo
  lat: number | null;         // approximate, used only for the schematic
route map
  lon: number | null;
  provisional: boolean;       // true only for the placeholder XXXX/XXX
}

type LegType = "landing" | "overflight" | "technical_stop" | "ferry";
interface Leg {
  id: string;
  flightNumber: string;       // prefix + number without suffix, e.g.
"PHC507"
  suffix: "" | "D";           // D = parallel flight; full number is
flightNumber + suffix
  aircraftId: string;
  backupAircraftId: string | null;
  legType: LegType;
  depIcao: string;
  arrIcao: string;
  etdUtc: IsoInstant;
  etaUtc: IsoInstant;
  status: "planned" | "cancelled";   // cancelled = CNL, see R7
  notes: string;
  source: { kind: "seed" | "import" | "manual"; sheetRef?: { sheet: string;
cell: string; rawText: string } };
}

interface Permit {
  id: string;
  legId: string;
  countryCode: CountryCode;
  type: "landing" | "overflight";
  number: string | null;
  status: "obtained" | "pending" | "cancelled";
  validFromUtc: IsoInstant | null;
  validToUtc: IsoInstant | null;
}
// Required permits per leg come from the route table in section 9. A
```

```
required permit with
// no Permit row is "missing" (R4).

interface MaintenanceBlock {
  id: string;
  aircraftId: string;
  kind: "aog" | "maintenance" | "c_check";
  startUtc: IsoInstant;
  endUtc: IsoInstant | null;  // null = open-ended
  note: string;
}

type RuleId = "R1" | "R2" | "R3" | "R4a" | "R4b" | "R5" | "R6";   // R7
produces no issue
interface ValidationIssue {
  id: string;                 // `${ruleId}:${sorted legIds}:${blockId ??
""}`, stable across recomputation
  ruleId: RuleId;
  severity: "error" | "warning";
  legIds: string[];
  aircraftId: string | null;
  blockId: string | null;
  whenUtc: IsoInstant;        // earliest ETD among legIds
  message: string;
  fixes: Fix[];
}
interface Fix {
  id: string;
  label: string;              // button text, e.g. "Reassign 2 legs to 4L-
PXA"
  enabled: boolean;
  disabledReason: string | null;   // shown as a tooltip and under the button
  patch: StatePatch;          // list of set / insert / remove operations,
applied atomically and undoable
}

// Issues are never stored: validate(state, demoClock) => ValidationIssue[]
is a pure function.
```

Derived fields (computed per leg, never stored)

```
const MIN = 60_000;
blockMinutes = (Date.parse(etaUtc) - Date.parse(etdUtc)) / MIN;
// valid when 0 < x < 1440
dayPlus      = utcDate(etaUtc) > utcDate(etdUtc) ? 1 : 0;
// printed as "(+1)"
arrLocal     = new Date(Date.parse(etaUtc) + arrAirport.utcOffsetMinutes *
MIN);   // read with getUTC* methods
dayTag       = localDate(arrLocal) !== utcDate(etaUtc) ?
isoWeekday(localDate(arrLocal)) : null;  // 1..7
```

- D1. The Excel sheet gives times of day only. The importer builds ETD from the row date and the ETD time. It builds ETA on the same date, and adds one day when the ETA time of day is not later than the ETD time of day (R6).
- D2. A leg is overnight when the UTC date of its ETA is later than the UTC date of its ETD. It belongs to the day of its ETD in every list, count and Schedule column. In the Gantt the bar continues across midnight into the next day.
- D3. "(+1)" is printed after the ETA, in UTC, when dayPlus is 1 (for example "01:30 (+1)"). Legs of 24 hours or more are not supported.
- D4. "DAY n LT" is printed on the arrival side when the local arrival date differs from the UTC arrival date. n is the ISO weekday (1 Monday to 7 Sunday) of the local arrival date. It is never printed for the placeholder airport. It is independent of (+1): a leg can show either, both or neither.
- D5. Display modes: "Dates" shows 14 Sep. "Days + dates + ISO weekday" shows Mon 14 Sep and the number 1.
- D6. Through-flight group: consecutive active legs on one aircraft, with the same full flight number and the same ETD date, where each departure station equals the previous arrival station and each ETD is not earlier than the previous ETA. The group id is the id of its first leg.
- D7. Every count (daily flights, KPIs, tab badges) excludes cancelled legs (R7).

Seed leg	ETD to ETA (UTC)	Block	(+1)	Local arrival	DAY n LT
PHC613 TBS to CCU, Fri 18 Sep	19:00 to 01:30	06:30	yes	Sat 07:00 (UTC+5:30)	none, same date as UTC arrival
PHC601 TBS to ALA, Mon 14 Sep	16:00 to 20:30	04:30	no	Tue 01:30 (UTC+5)	DAY 2 LT
PHC605 TBS to ALA, Thu 17 Sep	18:00 to 22:30	04:30	no	Fri 03:30 (UTC+5)	DAY 5 LT

Seed leg	ETD to ETA (UTC)	Block	(+1)	Local arrival	DAY n LT
PHC611 URC to XIY, Wed 16 Sep	09:30 to 12:30	03:00	no	Wed 20:30 (UTC+8)	none

5. Business rules

Seven rules run on every edit and on every import commit. Severity "error" means the operation as planned cannot work. Severity "warning" means a gap that needs attention but does not stop the leg.

Engine requirements

- E1. validate(state, demoClock) is a pure function. It returns issues sorted by severity (errors first), then whenUtc, then ruleId, then id.
- E2. Cancelled legs are excluded from every rule, as subject and as neighbour (R7). Legs touching the placeholder XXXX/XXX skip R2 and R4.
- E3. Issue granularity: R1 per full flight number per date, R2 per consecutive pair, R3 per aircraft and block, R4 per leg, R5 per pair.
- E4. A fix is one click, atomic and undoable (an Undo action stays in the toast). The list recomputes at once. A fix that would be unsafe is shown disabled with its reason.
- E5. The 48-hour window of R4 uses the demo clock, Sun 13 Sep 2026, 14:00Z, never the real clock.

ID	Rule	Severity	One-click fix
R1	Flight number conflict	error	Add suffix D to the later-departing leg
R2	Station continuity	error	Set the departure to the previous arrival
R3	Aircraft unavailable (AOG, maintenance, C-Check)	error	Reassign the legs to their backup aircraft, or cancel them
R4a	Permit urgent (departs within 48 h, a required permit is not obtained)	error	Mark the permits obtained

ID	Rule	Severity	One-click fix
R4b	Permit missing (a required permit has no record)	warning	Add pending permit rows
R5	Overlapping legs on one aircraft	error	Move the later leg to the previous ETA plus the minimum ground time
R6	Block time across midnight	error or warning	Open the leg (no automatic fix)
R7	Soft cancellation	no issue	Cancel leg and Restore leg

R1. Flight number conflict

Definition: group active legs by full flight number (flightNumber plus suffix) and by ETD UTC date. A group that holds more than one aircraft is an issue, unless its legs form a chain. A chain is a through-flight with an aircraft change: sorted by ETD, each departure station equals the previous arrival station and each ETD is not earlier than the previous ETA. Because the suffix is part of the full number, PHC507 and PHC507D never share a group, which is the D-suffix exemption. Legs of one aircraft never conflict.

Fix: "Make the 4L-PXB leg PHC509D" adds suffix D to the later-departing leg. It is disabled, with a reason, when that D number already exists on that date. Renumbering is done by hand in the leg drawer.

ID	Given	Expect
T-R1-1	PHC509 on 4L-PXA (05:00Z) and on 4L-PXB (06:00Z), Fri 18 Sep	1 error naming both legs. Fix label names 4L-PXB.
T-R1-2	PHC507 on 4L-PXA and PHC507D on 4L-PXB, Wed 16 Sep	No issue
T-R1-3	PHC611 TBS to URC and URC to XIY, both on 4L-PXC	No issue (one aircraft)
T-R1-4	PHC611 TBS to URC on 4L-PXC, PHC611 URC to XIY on 4L-PXD later the same day	No issue (chain with aircraft change)
T-R1-5	As T-R1-4, but the second leg departs TBS	1 error (not a chain)

ID	Given	Expect
T-R1-6	PHC509D on both aircraft, same date	1 error
T-R1-7	PHC509 on 4L-PXA on Fri 18 Sep and on 4L-PXB on Sat 19 Sep	No issue (different dates)
T-R1-8	The 4L-PXB PHC509 leg is cancelled	No issue
T-R1-9	Apply the fix on the seed	Issue gone, leg now reads PHC509D, no new issue

R2. Station continuity

Definition: per aircraft, order active legs by ETD. For each consecutive pair, when neither station is XXXX, it is an error if the previous arrival differs from the next departure. The first leg of an aircraft has no predecessor and is not checked. Cancelled legs are skipped when choosing the predecessor. The issue attaches to the next leg.

Fix: "Set departure to LIPK/FRL" sets the next leg's departure to the previous arrival and refreshes its permit rows. "Insert ferry leg" is offered as a secondary action that opens the leg drawer pre-filled.

ID	Given	Expect
T-R2-1	4L-PXB PHC511 arrives LIPK/FRL, then PHC512 departs LIMC/MXP (Thu 17 Sep)	1 error on PHC512, fix "Set departure to LIPK/FRL"
T-R2-2	PHC501 TBS to BEG, then PHC502 BEG to TBS	No issue
T-R2-3	Previous leg is the ferry PHC901 to XXXX/XXX, next leg departs TBS	No issue (placeholder)
T-R2-4	Previous leg is cancelled, the one before it arrived at TBS, next departs TBS	No issue
T-R2-5	First leg of an aircraft departs from any station	No issue

ID	Given	Expect
T-R2-6	Cancel PHC511 in the seed	R2 still fires on PHC512, now against UGTB/TBS, fix label "Set departure to UGTB/TBS"
T-R2-7	Apply the fix on the seed	Issue gone, PHC512 permit rows are those of FRL to TBS, no new issue

R3. Aircraft unavailable

Definition: a leg is affected when its interval [ETD, ETA) intersects a MaintenanceBlock [start, end) of its aircraft. A null end means open-ended. Touching boundaries are not an overlap. Setting an aircraft's registry status to AOG or Maintenance creates an open-ended block that starts at the demo clock. The affected legs of one block form one issue.

Fix A: "Reassign N legs to 4L-PXA" moves each leg to its own backupAircraftId. It is enabled only when every leg has a backup, no backup is inside a block that overlaps the leg, and no backup has an overlapping active leg. Otherwise it is disabled and names the first blocker. Fix B: "Cancel N legs (CNL)" is always enabled.

ID	Given	Expect
T-R3-1	Seed: AOG on 4L-PXB from Mon 21 Sep 20:00Z to Tue 22 Sep 10:00Z. PHC503 (04:00 to 07:30Z) and PHC504 (09:00 to 12:30Z) fly on Tue 22 Sep	1 error listing both legs (PHC504 overlaps for one hour)
T-R3-2	Apply Fix A on T-R3-1	Enabled, label "Reassign 2 legs to 4L-PXA". Afterwards no R3, R2 or R5 issue for these legs.
T-R3-3	Seed: C-Check on 4L-PXA from Sat 26 Sep 00:00Z to Sun 27 Sep 23:59Z. PHC513 (02:00Z) and PHC514 (08:30Z) fly on Sat 26 Sep	1 error. Fix A disabled: "4L-PXB flies PHC515 05:00 to 08:00 on Sat 26 Sep". Fix B enabled.
T-R3-4	A leg whose ETA equals the block start	No issue
T-R3-5	A leg that overlaps the block by 1 minute	Issue

ID	Given	Expect
T-R3-6	Cancel one of the two legs in T-R3-1, then the other	Issue lists 1 leg, then disappears
T-R3-7	Set 4L-PXC status to AOG at the demo clock	1 issue listing all 18 active 4L-PXC legs

R4. Permit risk

Definition: required permits for a leg come from the route table in section 9. A required permit is met only when a Permit row exists with status obtained. R4a (error): an active leg has ETD from the demo clock up to and including clock plus 48 hours, and at least one required permit is not met (pending, cancelled or no row). R4b (warning): an active leg later than that window has at least one required permit with no row. Pending permits beyond 48 hours raise no issue and appear on the Permits board only. One issue per leg lists the countries.

Fix for R4a: "Mark 2 permits obtained" sets the status, leaves the number empty (shown as "Not entered"), sets validity to the ETD date and the next day, and creates missing rows. Fix for R4b: "Add 2 pending permit rows".

ID	Given	Expect
T-R4-1	PHC501 Mon 14 Sep 03:30Z (13.5 h after the clock), Serbia landing and Romania overflight pending	1 R4a error
T-R4-2	PHC601 Mon 14 Sep 16:00Z (26 h), Kazakhstan landing pending	1 R4a error
T-R4-3	PHC611 Wed 16 Sep 02:00Z (60 h), permits obtained	No issue
T-R4-4	As T-R4-3 with one permit pending	No issue, the permit shows on the Permits board
T-R4-5	PHC613 Fri 18 Sep, India landing and UAE overflight have no rows	1 R4b warning listing both countries
T-R4-6	PHC517 Sun 20 Sep, Greece landing has no row	1 R4b warning

ID	Given	Expect
T-R4-7	A leg with ETD exactly Tue 15 Sep 14:00Z and a pending permit	R4a (the window is inclusive)
T-R4-8	ATH to TBS, which needs no third-country permit	No issue
T-R4-9	Ferry leg to XXXX/XXX	No issue
T-R4-10	A cancelled leg with a pending permit inside 48 h	No issue and not on the urgent list
T-R4-11	Apply the R4a fix on PHC501	Issue gone, two permits show Obtained with number "Not entered"

R5. Overlapping legs

Definition: per aircraft, order active legs by ETD. It is an error when the next ETD is earlier than the previous ETA. Equal values touch and are allowed. The issue names both legs.

Fix: "Move PHC510 to depart 08:30" sets the later leg's ETD to the previous ETA plus Company.minGroundMinutes (60) and keeps its block time.

ID	Given	Expect
T-R5-1	Seed: 4L-PXA PHC509 TBS to OTP 05:00 to 07:30Z, PHC510 OTP to TBS departs 07:00Z, Fri 25 Sep	1 error. Fix label "Move PHC510 to depart 08:30". After the fix PHC510 flies 08:30 to 11:00Z and no new issue appears.
T-R5-2	Next ETD equals previous ETA	No issue
T-R5-3	Next ETD is 1 minute before previous ETA	1 error
T-R5-4	Two aircraft with identical times	No R5
T-R5-5	PHC611 legs (ETA 07:30Z, next ETD 09:30Z)	No issue

ID	Given	Expect
T-R5-6	One of the overlapping legs is cancelled	No issue
T-R5-7	Leg A 23:00Z to 01:00Z (+1) and leg B 00:30Z on the next date, same aircraft	1 error

R6. Block time across midnight

Definition: the importer and the leg drawer take the ETA as a time of day. blockMinutes = (etaMinutes minus etdMinutes plus 1440) modulo 1440, where each value is minutes since 00:00 UTC. The ETA date is the ETD date, plus one day when the ETA time is not later than the ETD time. A block of 0 is invalid. A block under 1440 is valid, and above 1080 (18 h) it raises a warning. The drawer refuses to save an invalid leg and shows the message under the ETA field. Invalid legs can only arrive through import, where they are "needs review".

ID	ETD to ETA (UTC)	Expect
T-R6-1	03:30 to 06:30	180 min, "03:00", no (+1)
T-R6-2	19:00 to 01:30	390 min, "06:30", (+1)
T-R6-3	23:30 to 00:45	75 min, "01:15", (+1)
T-R6-4	23:59 to 00:00	1 min, "00:01", (+1)
T-R6-5	12:00 to 12:00	Invalid, "ETA equals ETD", not saved
T-R6-6	00:00 to 23:59	1439 min, valid, warning "longer than 18 h"
T-R6-7	Move the ETD date by one day	ETA moves with it, block unchanged
T-R6-8	ETD 10:00, ETA edited to 09:00	Read as next day: 1380 min, (+1), warning

R7. Soft cancellation excluded from counts

Definition: a leg with status cancelled (CNL) stays in the data and stays visible: struck through with a CNL tag on the Schedule, dashed on the Gantt, marked CNL in the Excel export. It is excluded from every count (Overview, daily flights, active legs), from R1 to R5 as subject and as neighbour, from the urgent and missing permit lists, and from checklist selection. Cancel is one click in the leg drawer and is reversible with Restore leg.

ID	Given	Expect
T-R7-1	Seed: 90 legs, 2 cancelled (PHC517 and PHC518, Sun 27 Sep)	Active legs 88. Sun 27 Sep reads 1 flight (PHC614), not 3.
T-R7-2	Cancel PHC503	The R3 issue for 4L-PXB lists 1 leg
T-R7-3	Cancel the 4L-PXB PHC509 duplicate	R1 issue gone
T-R7-4	Restore a cancelled leg	Counts and issues return
T-R7-5	Tick a cancelled leg in Checklist export	Checkbox disabled, tooltip "Cancelled legs cannot be added to a checklist"

6. Screen requirements

The screens below are those in the attached design board. Each criterion is written as Given, When, Then and refers to the full-demo seed in section 9 unless stated. The design copy that disagrees with this PRD is listed in section 9.

6.0 Shell and global behaviour

Top bar: AeroPlanner mark, operator name Phasis Air Cargo with "Fictional demo operator" and PHC / P7 / UGTB/TBS, an "ALL TIMES UTC" chip, the range selector (14 Sep to 4 Oct 2026, Weeks 1 to 3), theme toggle, help, and the user chip "OP Ops planner". Left navigation: Overview, Schedule, Conflicts, Permits, Checklist export, Fleet registry, Airport directory, Company profile, Import from Excel. Conflicts shows the open-issue count and Permits shows the urgent permit row count. A dismissible banner reads "Demo data. Fictional operator, all permit numbers are fake. Edits live in this session only."

ID	Given	When	Then
S0-1	The app is opened on a fresh tab	It finishes loading	Overview is shown, the banner is visible, the Conflicts badge reads 9, the Permits badge reads 3 and the range reads 14 Sep to 4 Oct 2026

ID	Given	When	Then
S0-2	The banner is visible	The user clicks Dismiss	It stays hidden for the tab session and returns after a reset
S0-3	Any screen	The user clicks Reset demo data	A dialog offers Full demo, Empty schedule and Cancel. Choosing either re-seeds in 1 s or less, opens Overview and shows a confirmation toast.
S0-4	Schedule is open	The user selects Week 2	Table and Gantt show 21 to 27 Sep only and the subtitle count follows. Overview and Conflicts are not filtered by the range.
S0-5	The URL carries presenter=1	The page loads	A list-state switcher (Normal, Loading, Error, Empty) appears. Without the parameter it is not rendered.
S0-6	Any screen	The user presses ?	A keyboard shortcut dialog opens and Esc closes it

6.1 Overview

ID	Given	When	Then
S1-1	Full demo	Overview opens	The subtitle reads "Week of 14 Sep 2026. Demo clock: Sun 13 Sep, 14:00Z." and four KPI tiles show: active legs 88, flights in the next 7 days 31, open issues 9, urgent permits 3
S1-2	Full demo	Overview opens	"Needs attention" shows "9 open" and the first 6 issues (level, type, issue, aircraft, when in UTC). "Show all 9 issues" opens Conflicts.
S1-3	A row in Needs attention	The user clicks Go to flight	Schedule opens on that leg's week, scrolls to it and highlights it for 2 s

ID	Given	When	Then
S1-4	No open issues	Overview opens	The list is replaced by "All clear. No open conflicts, urgent permits or continuity errors."
S1-5	Full demo	Overview opens	"Fleet availability, next 14 days" shows 4 aircraft rows for Sun 13 Sep to Sat 26 Sep. 4L-PXB is AOG on Mon 21 Sep and Tue 22 Sep, 4L-PXA is Maintenance on Sat 26 Sep, every other cell is Available. A day takes the worst state of any block touching it (AOG over Maintenance over Available).

6.2 Schedule: table view

Rows are days. Columns are aircraft, headed by registration and type read from the Fleet registry. Each leg shows flight number, route as IATA pair, times, "(+1)", "DAY n LT", permit state and CNL where they apply. A day row shows its flight count.

ID	Given	When	Then
S2-1	Week 1	Table view opens	7 day rows and 4 aircraft columns. Legs sit in ETD order. Day counts exclude cancelled legs. A leg with an open issue carries an issue marker.
S2-2	PHC611 on Wed 16 Sep	The table renders	Its two legs are joined by a double bar with a "Through" tag and carry no conflict marker
S2-3	PHC613 and PHC601	The table renders	PHC613 shows 19:00 to 01:30 (+1). PHC601 shows 16:00 to 20:30 and "DAY 2 LT".

ID	Given	When	Then
S2-4	Full demo	The user sets Status to "Has an issue"	Only legs with an open issue remain and the subtitle count follows. With no match the view shows "No legs match these filters" and a Clear filters action.
S2-5	Filters: aircraft type, station, flight number text, status (any, has an issue, permit pending or missing, permits obtained, cancelled)	Any combination is applied	Results update in 100 ms or less and Clear filters resets all
S2-6	The user ticks 3 legs	The selection bar appears	It reads "3 legs selected" with Clear and Open checklist preview. Preview opens Checklist export with those 3 ticked. Cancelled legs cannot be ticked.
S2-7	The Dates toggle	The user switches to "Days + dates + ISO weekday"	Day headers read like "Mon 14 Sep, 1"
S2-8	Add leg	The user clicks "+ Add leg"	The leg drawer opens empty (see 6.3)

6.2b Schedule: Gantt view

ID	Given	When	Then
S2-9	Gantt view	It opens	One lane per aircraft, hour ticks 00, 06, 12, 18, three weeks scrollable. AOG, maintenance and C-Check blocks are hatched. Overnight legs continue across midnight.

ID	Given	When	Then
S2-10	A leg is dragged to another aircraft lane or another time	The user drops it on free space	The leg updates (time snaps to 5 minutes, block time is kept), validation reruns and the issue list updates
S2-11	A leg is dragged	It is dropped on a hatched block	The drop is refused, the leg returns and a message reads "4L-PXB is AOG until Tue 22 Sep 10:00Z"
S2-12	A leg bar has focus	The user presses Enter	The leg drawer opens, so every drag has a keyboard route

6.3 Leg drawer (add and edit)

Fields: flight number with a D toggle, flight type (Landing, Overflight, Technical stop, Ferry), primary and backup aircraft, departure and arrival station, departure date (UTC), ETD and ETA in UTC, block time, arrival summary, permits for this route, special notes. Actions: Cancel this leg (CNL), Save leg, Cancel, Add to checklist.

ID	Given	When	Then
S3-1	A new leg	The user picks TBS as departure and ALA as arrival	The permit table shows Kazakhstan, Landing, Pending. Changing arrival to ATH replaces the rows with Greece, Landing. ATH to TBS shows "No third country permits needed for this route."
S3-2	ETD 19:00 and ETA 01:30	The user leaves the field	Block shows "06:30 +1" and the summary shows the arrival date and time in UTC and the local time at the destination
S3-3	ETA equals ETD	The user clicks Save leg	Save is blocked and "ETA must differ from ETD" appears under the field (R6)

ID	Given	When	Then
S3-4	A leg with an open issue	The drawer opens	The issue message and its fix buttons appear at the top. Clicking a fix applies it and the message becomes "No validation issues for this leg."
S3-5	An active leg	The user clicks Cancel this leg	Status becomes cancelled, the drawer closes, counts drop by one and a toast offers Undo
S3-6	Unsaved edits	The user presses Esc or Cancel	A "Discard changes?" prompt appears. With no edits the drawer closes at once.
S3-7	The flight number field	The user types a prefix other than PHC	The field shows "Prefix differs from company designator PHC" and still saves

6.4 Conflicts centre

Filters: severity, type, aircraft. Columns: severity, type, aircraft, date and time, issue with fix buttons, Go to flight. Side panels: "Rules checked" (R1 to R7 with definition and fix) and "Deliberately not flagged".

ID	Given	When	Then
S4-1	Full demo	The screen opens	9 issues: 7 errors then 2 warnings, each group ordered by date. The header reads "9 open".
S4-2	The R1 issue	The user clicks its fix	The issue leaves the list in 200 ms or less, a toast confirms with Undo, the nav badge reads 8 and Overview follows
S4-3	The R3 issue for 4L-PXA	The screen renders	"Reassign" is disabled with the visible reason "4L-PXB flies PHC515 05:00 to 08:00 on Sat 26 Sep" and "Cancel 2 legs (CNL)" is enabled

ID	Given	When	Then
S4-4	Type filter R4	It is applied	4 issues remain (2 errors, 2 warnings). A filter with no result shows "No conflicts in this view".
S4-5	Full demo	"Deliberately not flagged" renders	Two cards, generated from the engine's exemption hits: PHC611 through flight on Wed 16 Sep (TBS, URC, XIY on 4L-PXC) and PHC507 with PHC507D on Wed 16 Sep
S4-6	All issues fixed or none exist	The screen renders	"Clear. No conflicts in this view." with a Reset demo data action
S4-7	An issue row	The user clicks Go to flight	Same behaviour as S1-3

6.5 Permits board

ID	Given	When	Then
S5-1	Full demo	The screen opens	A pinned block "Urgent, departing within 48 hours" lists 3 rows by departure: PHC501 Serbia landing, PHC501 Romania overflight, PHC601 Kazakhstan landing
S5-2	Full demo	The grouped list renders	Country sections show these row counts: Romania 22, Serbia 18, Kazakhstan 12, China 9, UAE 8, Italy 6, Greece 5, Hungary 3, India 2. Total 85 rows, 53 obtained and 32 pending.
S5-3	The Country filter	"Kazakhstan" is chosen	Only Kazakhstan rows remain, pinned rows included
S5-4	A pending row	The user clicks Mark obtained	The status changes, the pinned block and the R4 issues recompute
S5-5	Cancelled legs	The board renders	They contribute no rows. Required permits with no row appear in Conflicts as R4b, not here.

ID	Given	When	Then
S5-6	A filter with no match	It is applied	"No permits match" with the hint "Try another status or country."

6.6 Checklist export

The preview is the document that gets exported (field list in section 8).

ID	Given	When	Then
S6-1	No legs ticked	The screen opens	Empty state "No legs selected" with the shortcut "Select the PHC611 through flight", which ticks both legs
S6-2	3 legs ticked	The ticks change	The preview rebuilds in 200 ms or less and the list header reads "3 selected" with Clear
S6-3	3 legs ticked	The user clicks Export PDF	An overlay "Building PDF file" shows a percentage. The file downloads within 3 s with the name pattern in section 8 and the overlay closes. Excel and Word behave the same.
S6-4	A ticked leg has an open error	The user exports	A confirmation reads "1 selected leg has open errors. Export anyway or review." The document itself carries no issue text.
S6-5	4L-PXD has no cargo payload	A 4L-PXD leg is ticked	The Payload cell reads "Missing" in the preview and in the file
S6-6	40 legs are ticked	The user ticks a 41st	The tick is refused with "A checklist holds up to 40 legs"

6.7 Fleet registry and aircraft drawer

ID	Given	When	Then
S7-1	Full demo	The screen opens	4 rows with tail, type, serial, year, MTOW, cargo payload and status. 4L-PXD carries an "Incomplete" badge because its payload is empty.
S7-2	The user clicks "+ Add aircraft"	They save valid data	The row appears, and a new column and a new Gantt lane exist at once. Registration must be unique and match 2 to 3 letters or digits, a dash and 3 letters (for example 4L-PXE).
S7-3	The registration of an aircraft is edited	They save	Every schedule header, Gantt lane, issue message and checklist shows the new value, because they read the registry
S7-4	Status is set to AOG or Maintenance	They save	An open-ended block starts at the demo clock and R3 runs (R3 definition, section 5)
S7-5	Any empty field	The row renders	The registry shows a warning and checklists print "Missing"

6.8 Airport directory

ID	Given	When	Then
S8-1	Full demo	The screen opens	14 rows (IATA, ICAO, airport, country, UTC offset). XXXX/XXX is tagged "Provisional (TBD)" with the note that it skips continuity and permit checks.
S8-2	The search box	The user types "mxp"	One row (LIMC/MXP). Matching is case-insensitive on IATA, ICAO and name.
S8-3	The search box	The user types "zzz"	"No airport matches" with a Clear search action

The directory is read only in the demo.

6.9 Company profile

ID	Given	When	Then
S9-1	Fields: legal name, VAT, ICAO designator, IATA designator, AOC number, base, address, operations contact, default checklist header text, logo	The user edits and clicks Save profile	A toast confirms and the next checklist preview shows the new header
S9-2	The logo control	The user uploads a PNG or SVG of 500 KB or less	The logo appears in the preview. Other types or sizes are rejected with a message. Without a logo the preview shows a "LOGO" placeholder.
S9-3	ICAO designator is not 3 letters, IATA is not 2 characters, or base is not in the directory	The user saves	Save is blocked with an inline message on that field

6.10 Import from Excel

Grammar, dirty variants and review handling are in section 7.

ID	Given	When	Then
S10-1	The screen opens	It renders	A drop zone, Browse files and "Use sample: Phasis_2026_Season.xlsx". A file that is not .xlsx is rejected with a message.

ID	Given	When	Then
S10-2	The sample is chosen	Parsing runs	A progress bar "Reading Phasis_2026_Season.xlsx" with a running count finishes in 3 s or less. Tiles read: legs parsed 90 from 118 rows and 1 sheet, need review 14, ready to import 76.
S10-3	The split view	It renders	Left: the raw cell text. Right: flight, route, times, permit state and note. A flagged row shows "Needs review", the problem and, where one exists, "Suggested: ..." with Accept fix.
S10-4	A flagged row with a suggestion	The user clicks Accept fix	The row shows "Fixed", need review drops by 1 and ready rises by 1
S10-5	A flagged row without a suggestion	The user clicks it	The leg drawer opens pre-filled with the raw text shown above the fields. Saving marks the row Fixed.
S10-6	Need review is 0	The user clicks "Import 90 legs"	Legs are written to the schedule, the app opens Schedule on Week 1 and a toast reads "90 legs imported". With rows unresolved the button reads "Import 76 legs" and the toast adds "14 rows skipped".
S10-7	Import is in progress or done	The user clicks Start over	The screen returns to the drop zone and nothing is imported

6.11 List states

Every list (Schedule, Conflicts, Permits, Fleet, Airports) has loading (skeleton rows), empty and error states. The error state reads "Could not load this list. The planning service did not respond (demo error, code 503). Your edits are safe." with Try again. In the demo these states appear only through the presenter switcher (S0-5).

7. Excel import

The importer reads one season sheet, normalises each cell line, parses it against a small grammar, and sends anything it cannot trust to "needs review" with a suggestion where one is safe. Parsing runs in the browser. Nothing is written to the schedule until the user clicks Import.

7.1 Workbook layout the parser expects

This layout is an assumption taken from the design and the example string. It must be checked against real sheets (open question 11).

- One sheet per season. A title line above the grid is ignored.
- Aircraft header row: one cell per aircraft holding the registration (4L-PXA), optionally with the type on a second line. Header cells may be merged across columns.
- Column A holds date labels: "Mon 14.09", "14/09/2026", "14 Sep" or a real Excel date. The year comes from the nearest preceding week banner or the sheet name. If no year can be found, the line needs review.
- Week banner rows ("WEEK 1, 14-20 SEP 2026") are merged across the grid and treated as structure.
- A grid cell (date row by aircraft column) holds zero or more leg lines separated by line breaks. Each line is one leg of that aircraft that UTC date.
- A merged range is read once, from its top-left cell, and applies to every cell it covers. A merged leg cell that spans several aircraft columns is read for the leftmost aircraft only.
- Colour codes on a leg cell: green fill means permits obtained, amber fill means pending, red fill or strikethrough means cancelled.

The sample workbook has 118 "rows", counting a row as one non-empty cell line: 90 leg lines and 28 structure lines (4 aircraft headers, 3 week banners, 21 date labels). The design text "517 legs parsed from 812 rows" must change to match.

7.2 Normalisation, in this order

1. Unicode NFKC. Replace non-breaking spaces, tabs and zero-width characters with a space. Collapse runs of whitespace and trim.
2. Upper-case everything except the free note.
3. Remove spaces around slashes: "UGTB / TBS" becomes "UGTB/TBS".
4. Time tokens: accept ETD, ETA or STD, a separator of "/", ":" or space, a value HH.MM, HH:MM or HHMM, and an optional Z. Rewrite as "ETD/HH.MMZ" (STD becomes ETD).

5. Split glued markers: "06.30Z+" becomes "06.30Z +".
6. Day tags: "DAY4 LT", "DAY 4 LT", "(DAY 4 LT)" and "(DAY4LT)" become "DAY 4 LT".
7. Flight numbers: "CMS572 D" and "CMS 572D" become "CMS572D".

7.3 Grammar

```
line        = flight , stationTime , stationTime , { tail } ;
flight      = prefix , number , [ suffix ] ;        (* prefix: 3 letters,
number: 1 to 4 digits, suffix: 1 letter *)
stationTime = station , timeTok ;
station     = icao , "/" , iata ;                   (* icao: 4 letters, iata:
3 letters *)
timeTok     = ( "ETD" | "ETA" ) , "/" , hh , "." , mm , [ "Z" ] ;
tail        = dayTag | marker | "CNL" | typeWord | note ;
dayTag      = "DAY" , digit17 , "LT" ;              (* digit17: 1 to 7 *)
marker      = "+" | "-" ;
typeWord    = "FERRY" | "OVERFLIGHT" | "TECHSTOP" ;
note        = any text up to the end of the line, kept verbatim in its
original case ;
```

- The first stationTime is the departure and the second is the arrival. After them, tail items may come in any order. The note starts at the first token that is not a dayTag, marker, CNL or typeWord.
- A marker is a standalone "+" or "-" that directly follows the arrival time or a dayTag. A "-" or "+" anywhere else is part of the note.
- legType is ferry when typeWord is FERRY or either station is XXXX/XXX, overflight or technical_stop for the other words, otherwise landing.
- ETD date is the row date. ETA date follows R6 (same date, or the next day when the ETA time is not later than the ETD time).
- Permit markers (assumed, open question 1): "+" creates every required permit row with status obtained and no number. "-" creates them pending. No marker creates no rows, which raises R4b. Precedence for status: CNL, strikethrough or red fill (cancelled) over marker over green or amber fill over nothing.
- The sheet's "DAY n" must equal the ISO weekday of the local arrival date. The app prints the tag only where the local date differs from the UTC date (D4).

7.4 Dirty variants and how each is handled

Variant	Example	Handling	Flagged
Stray spaces	`CMS572D UGTB / TBS ETD/ 03.30Z`	Normalised	No
Time without Z, with colon or no separator	`ETD/03:30`, `ETD/0330Z`	Normalised	No
Lower case	`etd/03.30z`	Normalised	No
ETA typed instead of ETD	`UGTB/TBS ETA/03.30Z LYBE/BEG ETA/06.30Z`	First time is read as the departure. Code ETA_FOR_ETD. Suggestion: change the first tag to ETD. The mirror case (two ETD) gets ETD_FOR_ETA and suggests changing the second tag.	Yes
"DAY4 LT"	`... ETA/20.30Z DAY4 LT`	Normalised to DAY 4 LT. Compared with the ISO weekday of the local arrival date.	Only on mismatch (DAYTAG_MISMATCH, suggestion: the derived number)
"(DAY 4 LT)"	`... ETA/22.30Z (DAY 4 LT)`	As above	Only on mismatch
Trailing "+" or "-"	`... ETA/06.30Z +`, `ETA/06.30Z+`	Read as the permit marker	No
Both markers	`... ETA/07.30Z + -`	Code MARKER_CONFLICT, no suggestion	Yes
Free notes	`... deliver to dock 4`, `TBC`	Kept verbatim in Leg.notes	No

Variant	Example	Handling	Flagged
Note that contradicts a time	`... ETA/01.30Z ETD 19.30Z TBC`	Code NOTE_TIME_CONFLICT. Suggestion: keep the structured ETD and keep the text as a note.	Yes
Foreign flight prefix	`CMS572D ...` when the company designator is PHC	Code PREFIX_MISMATCH. Suggestion: replace the prefix (PHC572D).	Yes
Station pair does not match the directory	`LROP/BUH`, `LIMC/MIL`	The directory is looked up by ICAO. Code STATION_PAIR_MISMATCH, suggestion: the directory pair (LROP/OTP, LIMC/MXP). An unknown ICAO gets STATION_UNKNOWN and no suggestion.	Yes
Malformed or impossible time	`ETD/09.3Z`, `ETD/25.00Z`	Codes TIME_MALFORMED and TIME_OUT_OF_RANGE, no suggestion	Yes
ETA equals ETD	`ETD/01.00Z ... ETA/01.00Z`	Code BLOCK_ZERO (R6), no suggestion	Yes
Same departure and arrival	`OMDW/DWC ... OMDW/DWC`	Code SAME_STATION, no suggestion	Yes
Placeholder destination	`XXXX/XXX`	Ferry leg, skips R2 and R4	No
CNL	`... CNL`	Leg is cancelled (R7)	No
Line that matches no flight number	`STANDBY CREW`	Code UNPARSEABLE_LINE, no suggestion	Yes

Variant	Example	Handling	Flagged
Aircraft header not in the registry	`4L-PXE`	Whole column flagged AIRCRAFT_UNKNOWN. Suggestion: add it to the Fleet registry as incomplete.	Yes

When a line carries several flags, only the first is shown, in this precedence: structure, time, station, prefix, marker, day tag, note. The row counts once. The user fixes one flag at a time and the line is parsed again after each fix.

7.5 Parser test cases

Each line sits in the cell for Mon 14 Sep under the stated aircraft unless noted.

ID	Input (company ICAO in brackets)	Expected
P1	`CMS572D UGTB/TBS ETD/03.30Z LYBE/BEG ETA/06.30Z +` (CMS)	flightNumber CMS572, suffix D, UGTB to LYBE, ETD 2026-09-14T03:30Z, ETA 06:30Z, block 03:00, permits obtained, no flags
P2	The same line (PHC)	The same leg, flag PREFIX_MISMATCH, suggestion `PHC572D`
P3	`cms572d ugtb / tbs etd/ 03:30 lybe/beg eta/0630z+` (CMS)	Identical to P1, no flags
P4	`PHC601 UGTB/TBS ETD/16.00Z UAAA/ALA ETA/20.30Z DAY4 LT -`	Flag DAYTAG_MISMATCH, suggestion "DAY 2 LT", marker pending
P5	`PHC601 UGTB/TBS ETD/16.00Z UAAA/ALA ETA/20.30Z (DAY 2 LT) -`	No flags
P6	`PHC613 UGTB/TBS ETD/19.00Z VECC/CCU ETA/01.30Z +`	Block 06:30, ETA 2026-09-15T01:30Z shown as (+1), no flags
P7	`PHC901 UGTB/TBS ETD/08.00Z XXXX/XXX ETA/12.00Z FERRY`	Leg type ferry, no permit rows, no flags
P8	`PHC513 UGTB/TBS ETD/02.00Z LIMC/MXP ETA/06.30Z CNL`	Cancelled leg, no flags

ID	Input (company ICAO in brackets)	Expected
P9	`PHC507D UGTB/TBS ETD/04.00Z LGAV/ATH ETA/07.00Z + Deliver to dock 4`	Suffix D, note "Deliver to dock 4" in original case
P10	`PHC512 LIMC/MXP ETD/09.00Z UGTB/TBS ETA/13.00Z`	No marker, no permit rows, no parse flag (R4b and R2 are raised after import)
P11	`PHC614 VECC/CCU ETD/20.30Z UGTB/TBS ETA/03.00Z +`	ETA on the next date, block 06:30
P12	`PHC606 UAAA/ALA ETD/01.00Z UGTB/TBS ETA/01.00Z +`	Flag BLOCK_ZERO

7.6 Needs-review handling

- RV1. Every flag creates a ReviewItem: id, sheetRef (sheet, cell, line), rawText, code, message and suggestion (or null). A flagged line is held in memory with best-effort fields and is not importable until resolved.
- RV2. A flagged row shows "Needs review", the problem and, when a suggestion exists, "Suggested: ..." and Accept fix. Without a suggestion it offers Edit, which opens the leg drawer with the raw text above the fields.
- RV3. Accept fix changes the parsed fields, never the raw text, then reruns parsing and R6. If a new flag appears, the row stays flagged with the new reason. Otherwise it shows "Fixed".
- RV4. Counters: Need review is the number of unresolved flagged rows. Ready to import is unflagged rows plus fixed rows. Parsed is their sum.
- RV5. Import writes unflagged and fixed legs. Unresolved rows stay listed and are skipped; the toast says how many.
- RV6. Idempotence: the natural key is full flight number, ETD instant, departure ICAO and aircraft. A leg whose key exists is updated, otherwise inserted. Nothing is deleted. Importing the sample twice leaves 90 legs.
- RV7. Provenance: Leg.source keeps sheet, A1-style cell (with "line 2" for multi-line cells) and rawText. The leg drawer shows "Imported from B7" with the raw text.
- RV8. State lives only in memory. Leaving the screen or clicking Start over discards the parse.

7.7 The 14 flagged cells in the sample workbook

All fourteen are in different cells and each carries exactly one flag. After resolution the 90 imported legs equal the seed legs of section 9 on the natural key; permit rows carry statuses from the markers and no numbers.

#	Leg and date	Aircraft	Raw text as typed	Flag	Suggestion	Correct value
1	PHC502, Mon 14 Sep	4L-PXA	`PHC502 LYBE/BEG ETA/08.00Z UGTB/TBS ETA/11.00Z +`	ETA_FOR_ETD	ETD/08.00Z	ETD 08:00
2	PHC515, Sat 19 Sep	4L-PXB	`PHC515 UGTB/TBS ETA/05.00Z LYBE/BEG ETA/08.00Z +`	ETA_FOR_ETD	ETD/05.00Z	ETD 05:00
3	PHC601, Mon 14 Sep	4L-PXC	`PHC601 UGTB/TBS ETD/16.00Z UAAA/ALA ETA/20.30Z DAY 3 LT -`	DAYTAG_MISMATCH	DAY 2 LT	DAY 2 LT
4	PHC605, Thu 17 Sep	4L-PXD	`PHC605 UGTB/TBS ETD/18.00Z UAAA/ALA ETA/22.30Z (DAY 4 LT) +`	DAYTAG_MISMATCH	DAY 5 LT	DAY 5 LT
5	PHC504, Tue 15 Sep	4L-PXB	`PHC504 LHBP/BUD ETD/09.3Z UGTB/TBS ETA/12.30Z +`	TIME_MALFORMED	none	ETD 09:00
6	PHC612 first leg, Thu 17 Sep	4L-PXC	`PHC612 ZLXY/XIY ETD/25.00Z ZWWW/URC ETA/05.00Z +`	TIME_OUT_OF_RANGE	none	ETD 02:00

#	Leg and date	Aircraft	Raw text as typed	Flag	Suggestion	Correct value
7	PHC510, Fri 18 Sep	4L-PXA	`PHC510 LROP/BUH ETD/09.00Z UGTB/TBS ETA/11.30Z +`	STATION_PAIR_MISMATCH	LROP/OTP	LROP/OTP
8	PHC514, Sat 19 Sep	4L-PXA	`PHC514 LIMC/MIL ETD/08.30Z UGTB/TBS ETA/13.00Z +`	STATION_PAIR_MISMATCH	LIMC/MXP	LIMC/MXP
9	PHC516, Sat 19 Sep	4L-PXB	`CMS516 LYBE/BEG ETD/09.30Z UGTB/TBS ETA/12.30Z +`	PREFIX_MISMATCH	PHC516	PHC516
10	PHC518, Sun 20 Sep	4L-PXB	`CMS518 LGAV/ATH ETD/09.00Z UGTB/TBS ETA/12.00Z +`	PREFIX_MISMATCH	PHC518	PHC518
11	PHC606, Fri 18 Sep	4L-PXD	`PHC606 UAAA/ALA ETD/01.00Z UGTB/TBS ETA/01.00Z +`	BLOCK_ZERO	none	ETA 05:30
12	PHC503, Tue 15 Sep	4L-PXB	`PHC503 UGTB/TBS ETD/04.00Z LHBP/BUD ETA/07.30Z + -`	MARKER_CONFLICT	none	marker "+"
13	PHC613, Fri 18 Sep	4L-PXD	`PHC613 UGTB/TBS ETD/19.00Z VECC/CCU ETA/01.30Z ETD 19.30Z TBC`	NOTE_TIME_CONFLICT	keep ETD 19.00Z, keep note	ETD 19:00
14	PHC604, Tue 15 Sep	4L-PXD	`PHC604 OMDW/DWC ETD/08.00Z OMDW/DWC ETA/12.00Z +`	SAME_STATION	none	arrival UGTB/TBS

Rows 1 to 4, 7 to 10 and 13 carry a suggestion (9 rows). Rows 5, 6, 11, 12 and 14 do not (5 rows) and are fixed in the leg drawer.

The sample also holds 18 cells that are cleaned silently and never flagged: 6 with stray spaces, 4 with a colon, no Z or no separator in a time, 3 in lower case, 3 with a consistent "DAY n LT" typed without a space (PHC601 and PHC605 in week 2, PHC611 URC to XIY in week 2) and 2 in the form "(DAY 2 LT)" or "(DAY 5 LT)" (PHC601 and PHC605 in week 3). The exact cells are fixed in the fixture file and the counts are asserted by test. Four further cells hold free notes that are kept without a flag (PHC611 and PHC507D in week 1, PHC601 and PHC901 in week 2).

8. Checklist export

One selection of legs produces one consolidated document in PDF, Excel or Word. All three are built from one layout model, so a block appears in the same order with the same content in each format and in the on-screen preview.

8.1 Document content, in order

Block	Fields	Source and rules
1. Header	Logo (or a "LOGO" placeholder box), legal name, line "ICAO PHC | IATA P7 | AOC DEMO-0001 | Base UGTB/TBS", default header text	Company profile
2. Title	"FLIGHT CHECKLIST" and the date span	Span from the earliest to the latest ETD date. Formats: "16 Sep 2026", "16 to 17 Sep 2026", "30 Sep to 2 Oct 2026", "30 Dec 2026 to 2 Jan 2027".
3. Aircraft	Role, registration, type, serial no., year, MTOW, payload	One row per distinct aircraft. Primary rows (aircraft flying a selected leg) first, then backup rows for backups that are not already primary. MTOW and payload print as "79,016 kg". An empty field prints "Missing".

Block	Fields	Source and rules
4. Route	Schematic map with a line per leg in ETD order, IATA labels at stations and a legend	Built from the Airport directory coordinates. Straight lines, no map tiles, no network request. Labelled "schematic, not to scale".
5. Legs	Flight, aircraft, route (IATA to IATA), date, ETD, ETA, block, type	Sorted by ETD. Flight shows the full number with suffix. ETA carries "(+1)" when dayPlus is 1. "DAY n LT" is printed under the ETA where D4 applies. Times are UTC, header says "ETD (UTC)".
6. Permits per country	Country, permit, flight, number, status, valid	One row per Permit per selected leg, sorted by country, then ETD. A required permit with no row prints "No record" in the Status cell. A null number prints "Not entered". Valid prints as "16 to 17 Sep".
7. Special notes	Each non-empty leg note, prefixed with its flight number	Leg.notes. If none, the block reads "None".
8. Signatures	"Prepared by, Flight Operations" and "Commander signature and date"	Two empty signature lines with a name and date caption
9. Footer	"All times UTC unless marked LT. Demo document, fictional operator.", page x of y, generated timestamp	The timestamp uses the demo clock so repeated runs match

Layout: A4 portrait, 15 mm margins, one sans-serif font embedded in the PDF with Latin coverage. Table header rows repeat on each page, rows never split across pages, and the signature block stays together on the last page.

8.2 Batch behaviour

- B1. One document per export, not one per leg. The selection holds 1 to 40 legs and may mix aircraft and dates.

- B2. Ticks come from the Schedule, the Checklist export list and "Add to checklist" in the leg drawer. The selection persists across screens for the session and is cleared by Reset.
- B3. Cancelled legs cannot be ticked (R7). Legs with open errors can be exported after the confirmation in S6-4, and the document shows no issue text.
- B4. File name: `PHC_Checklist_<first ETD date>[_to_<last ETD date>]_<n>leg(s).<ext>` with ISO dates, for example `PHC_Checklist_2026-09-16_to_2026-09-17_3legs.pdf` and `PHC_Checklist_2026-09-18_1leg.pdf`. The prefix is the company ICAO designator.
- B5. The export overlay reads "Building PDF file" (or Excel, Word) with a percentage. On failure it reads "Could not build the file. Try again." and keeps the selection.
- B6. Libraries load on first use, so the initial bundle is unaffected (section 10).
- B7. Optional, not needed for the pitch (priority P2): "One file per aircraft" downloads a ZIP with one document per aircraft.

8.3 Format specifics

- PDF: vector text that can be selected and searched, fonts embedded, document title and author set ("Phasis Air Cargo (demo)"). Target size under 300 KB for 3 legs.
- Excel (.xlsx): sheet "Checklist" holds the header, aircraft table, legs table and notes. Sheet "Permits" holds the permit table. Times are real Excel times formatted hh:mm with "(UTC)" in the header, dates are real dates, an "ETA day" column holds "+1" or stays empty. Header rows are frozen, A4 landscape, fit to one page wide.
- Word (.docx): the same blocks as editable text and native tables. The route map is an embedded image. Page numbers use a field, so they update when the user edits.

8.4 Acceptance tests

ID	Given	Then
X1	Storyline C: PHC611 TBS to URC (Wed 16 Sep 02:00 to 07:30Z, block 05:30), PHC611 URC to XIY (09:30 to 12:30Z, block 03:00), PHC612 XIY to URC (Thu 17 Sep 02:00 to 05:00Z, block 03:00)	PDF title span "16 to 17 Sep 2026". Aircraft table: Primary 4L-PXC and Backup 4L-PXD, whose payload reads "Missing". 3 legs rows. 4 permit rows: China landing (3 legs) and Kazakhstan overflight (PHC611 TBS to URC), all Obtained. Notes block: "PHC611: Confirm handling agent at URC". File name as in B4.

ID	Given	Then
X2	One leg: PHC613 TBS to CCU, Fri 18 Sep	ETA reads "01:30 (+1)" and block "06:30". The permits table shows "No record" for India landing and UAE overflight. Aircraft table: Primary 4L-PXD and Backup 4L-PXC. Notes block: "PHC613: ETD 19.30Z TBC".
X3	Excel export of X1	2 sheets, 3 legs rows, ETD and ETA are real time values, header says UTC
X4	Word export of X1	Opens in Word with the same block order as the PDF, tables are editable
X5	14 legs selected	The legs table runs onto page 2 with its header repeated and the signature block is not split
X6	A logo is uploaded in Company profile	It appears in the preview and in all three formats
X7	A cancelled leg exists in the range	It is absent from the list's ticks and from every format
X8	12 legs, then 40 legs	PDF build takes 3 s or less, then 8 s or less, on the reference laptop (section 10)
X9	Preview text versus exported PDF text for X1	Same blocks in the same order with the same values

9. Seed data

The fictional airline Phasis Air Cargo has 4 aircraft and 90 legs over 3 weeks (Mon 14 Sep to Sun 4 Oct 2026), generated from one weekly pattern plus a short list of planted exceptions. The totals in 9.8 are asserted by test.

Principles. No randomness and no real clock: the demo clock is 2026-09-13T14:00:00Z. Seed files are committed fixtures (`seed/*.ts`, plus the sample workbook `Phasis_2026_Season.xlsx` built from the same data). Leg ids are `L-<yyyymmdd>-<full flight number>-<dep IATA>-<aircraft letter>`, for example `L-20260918-PHC509-TBS-B`. All names, serial numbers and permit numbers are fictional.

9.1 Company

Legal name Phasis Air Cargo, ICAO PHC, IATA P7, AOC DEMO-0001, VAT 000000000 (demo), base UGTB, address "Tbilisi, Georgia (fictional)", operations contact ops@phasis-demo.example, default checklist header text "Flight packet for crew and handling agents", no logo, minimum ground time 60 minutes.

9.2 Fleet

Registration	Type	Serial	Year	MTOW (kg)	Cargo payload (kg)	Status	Default backup
4L-PXA	B737-800SF	34001	2006	79,016	23,000	active	4L-PXB
4L-PXB	B737-800BCF	34002	2004	79,016	22,000	active	4L-PXA
4L-PXC	B767-300ER	29001	1999	186,880	50,000	active	4L-PXD
4L-PXD	B767-300ER	29002	2001	186,880	empty on purpose (shows "Incomplete" and "Missing")	active	4L-PXC

Every leg's backup aircraft is the default backup of its primary aircraft.

9.3 Airports

Offsets are fixed and valid for the demo range, because European summer time ends on 25 Oct 2026. Coordinates are approximate and used only for the schematic route map.

ICAO	IATA	Airport	Country	UTC offset	Latitude, longitude
UGTB	TBS	Tbilisi International	GE	+4:00	41.669, 44.955
ZWWW	URC	Urumqi Diwopu	CN	+8:00	43.907, 87.474
ZLXY	XIY	Xi'an Xianyang	CN	+8:00	34.447, 108.752
UAAA	ALA	Almaty International	KZ	+5:00	43.352, 77.040
LYBE	BEG	Belgrade Nikola Tesla	RS	+2:00	44.818, 20.309
LGAV	ATH	Athens Eleftherios Venizelos	GR	+3:00	37.936, 23.944
LHBP	BUD	Budapest Ferenc Liszt	HU	+2:00	47.437, 19.262
LROP	OTP	Bucharest Henri Coanda	RO	+3:00	44.572, 26.102
LIMC	MXP	Milan Malpensa	IT	+2:00	45.630, 8.723

ICAO	IATA	Airport	Country	UTC offset	Latitude, longitude
VHHH	HKG	Hong Kong International	HK	+8:00	22.309, 113.915
VECC	CCU	Kolkata Netaji Subhas Chandra Bose	IN	+5:30	22.654, 88.447
LIPK	FRL	Forli	IT	+2:00	44.195, 12.070
OMDW	DWC	Dubai Al Maktoum International	AE	+4:00	24.896, 55.161
XXXX	XXX	Placeholder (TBD)	XX	+0:00	none, never drawn

HKG has no seeded leg. It stays in the directory and in the station filter, as in the design.

9.4 Required permits by route

Route	Flying from TBS needs	Flying to TBS needs
TBS and BEG	Serbia landing, Romania overflight	Romania overflight
TBS and ATH	Greece landing	none
TBS and BUD	Hungary landing, Romania overflight	Romania overflight
TBS and OTP	Romania landing	none
TBS and MXP	Italy landing, Serbia overflight	Serbia overflight
TBS and FRL	Italy landing, Serbia overflight	Serbia overflight
TBS and URC	China landing, Kazakhstan overflight	Kazakhstan overflight
URC and XIY (either direction)	China landing	China landing
TBS and ALA	Kazakhstan landing	none
TBS and DWC	UAE landing	none
TBS and CCU	India landing, UAE overflight	UAE overflight
Any leg touching XXXX/XXX	none	none

9.5 Weekly pattern (28 legs, repeated in weeks 1, 2 and 3)

Day 1 of week 1 is Mon 14 Sep, of week 2 Mon 21 Sep, of week 3 Mon 28 Sep. Every leg is type landing. Times are UTC.

Day	Aircraft	Flight	Route	ETD	ETA	Block
Mon	4L-PXA	PHC501	TBS to BEG	03:30	06:30	03:00
Mon	4L-PXA	PHC502	BEG to TBS	08:00	11:00	03:00
Mon	4L-PXC	PHC601	TBS to ALA	16:00	20:30	04:30
Tue	4L-PXC	PHC602	ALA to TBS	02:00	06:30	04:30
Tue	4L-PXD	PHC603	TBS to DWC	02:00	06:00	04:00
Tue	4L-PXB	PHC503	TBS to BUD	04:00	07:30	03:30
Tue	4L-PXD	PHC604	DWC to TBS	08:00	12:00	04:00
Tue	4L-PXB	PHC504	BUD to TBS	09:00	12:30	03:30
Wed	4L-PXC	PHC611	TBS to URC	02:00	07:30	05:30
Wed	4L-PXA	PHC507	TBS to ATH	04:00	07:00	03:00
Wed	4L-PXA	PHC508	ATH to TBS	08:30	11:30	03:00
Wed	4L-PXC	PHC611	URC to XIY	09:30	12:30	03:00
Thu	4L-PXC	PHC612	XIY to URC	02:00	05:00	03:00
Thu	4L-PXB	PHC511	TBS to FRL	03:00	07:00	04:00
Thu	4L-PXC	PHC612	URC to TBS	07:00	12:30	05:30
Thu	4L-PXB	PHC512	FRL to TBS	09:00	13:00	04:00
Thu	4L-PXD	PHC605	TBS to ALA	18:00	22:30	04:30
Fri	4L-PXD	PHC606	ALA to TBS	01:00	05:30	04:30
Fri	4L-PXA	PHC509	TBS to OTP	05:00	07:30	02:30
Fri	4L-PXA	PHC510	OTP to TBS	09:00	11:30	02:30
Fri	4L-PXD	PHC613	TBS to CCU	19:00	01:30 (+1)	06:30
Sat	4L-PXA	PHC513	TBS to MXP	02:00	06:30	04:30
Sat	4L-PXB	PHC515	TBS to BEG	05:00	08:00	03:00
Sat	4L-PXA	PHC514	MXP to TBS	08:30	13:00	04:30
Sat	4L-PXB	PHC516	BEG to TBS	09:30	12:30	03:00
Sun	4L-PXB	PHC517	TBS to ATH	04:30	07:30	03:00
Sun	4L-PXB	PHC518	ATH to TBS	09:00	12:00	03:00

Day	Aircraft	Flight	Route	ETD	ETA	Block
Sun	4L-PXD	PHC614	CCU to TBS	20:30	03:00 (+1)	06:30

9.6 Planted exceptions

Six extra legs (84 pattern legs plus 6 gives 90):

Date	Aircraft	Flight	Route	ETD to ETA	Type	Purpose
Wed 16 Sep	4L-PXB	PHC507D	TBS to ATH	04:00 to 07:00	landing	Parallel flight with PHC507, must not raise R1
Wed 16 Sep	4L-PXB	PHC508D	ATH to TBS	08:30 to 11:30	landing	Return of the above
Fri 18 Sep	4L-PXB	PHC509	TBS to OTP	06:00 to 08:30	landing	Duplicate of the 4L-PXA PHC509, raises R1
Fri 18 Sep	4L-PXB	PHC520	OTP to TBS	10:00 to 12:30	landing	Return of the duplicate, so the aircraft ends at TBS and R2 stays quiet
Wed 23 Sep	4L-PXD	PHC901	TBS to XXXX/XXX	08:00 to 12:00	ferry	Destination not decided, skips R2 and R4
Thu 24 Sep	4L-PXD	PHC902	XXXX/XXX to TBS	08:00 to 12:00	ferry	Return from the placeholder

Changes to pattern legs

Leg	Change	Raises
PHC512, Thu 17 Sep (4L-PXB)	Departs LIMC/MXP instead of LIPK/FRL	R2
PHC510, Fri 25 Sep (4L-PXA)	ETD 07:00, ETA 09:30 instead of 09:00 and 11:30	R5 against PHC509 (ETA 07:30)
PHC517 and PHC518, Sun 27 Sep (4L-PXB)	Status cancelled	R7, no issue

Maintenance blocks

Aircraft	Kind	From	To	Raises
4L-PXB	AOG	Mon 21 Sep 20:00Z	Tue 22 Sep 10:00Z	R3 on PHC503 and PHC504 (Tue 22 Sep)
4L-PXA	C-Check	Sat 26 Sep 00:00Z	Sun 27 Sep 23:59Z	R3 on PHC513 and PHC514 (Sat 26 Sep)

Permit rows. Rows are created from the route table in 9.4. Weeks 1 and 2 legs are obtained, except PHC501 and PHC601 on Mon 14 Sep (all their rows pending). All week 3 legs are pending. PHC613 and PHC517 on week 1 have no rows at all (R4b). Obtained rows get the number `DEMO-<country code>-<10000 + n>`, n counting obtained rows in ETD order. Pending rows have no number. Validity runs from 00:00Z on the ETD date to 23:59Z on the next day. The cancelled PHC517 on Sun 27 Sep keeps one Greece landing row with status cancelled, which the board hides.

9.7 Notes on the seed

The design's "Not flagged" cards (PHC611 through flight and PHC507 with PHC507D on Wed 16 Sep) and its 12 station filter entries match this seed. Leg notes in the seed: PHC611 TBS to URC on Wed 16 Sep "Confirm handling agent at URC", PHC507D on Wed 16 Sep "Deliver to dock 4", PHC613 on Fri 18 Sep "ETD 19.30Z TBC", PHC601 on Mon 21 Sep "Cargo: perishables" and PHC901 on Wed 23 Sep "Destination to be decided". All other legs have no note.

9.8 Expected totals (asserted by test)

Measure	Value
Legs, active, cancelled	90, 88, 2
Legs per week	32, 30, 28
Active legs per aircraft	4L-PXA 24, 4L-PXB 26, 4L-PXC 18, 4L-PXD 20
Flights in the next 7 days from the demo clock	31
Through-flight groups	6 (PHC611 and PHC612, each in 3 weeks)
Overnight legs (+1)	6 (PHC613 and PHC614, each in 3 weeks)

Measure	Value
"DAY n LT" tags	6 (PHC601 shows DAY 2, PHC605 shows DAY 5, each in 3 weeks)
Permit rows shown on the board	85 (53 obtained, 32 pending)
Urgent permit rows	3
Open issues	9: R1 1, R2 1, R3 2, R4a 2, R4b 2, R5 1 (7 errors, 2 warnings)
Issues after the three storyline B fixes	6
Fleet rows, incomplete rows	4, 1
Airports	14

9.9 Design copy to change

In the design	Change to
"517 legs parsed from 812 rows, 1 sheet"	"90 legs parsed from 118 rows, 1 sheet"
The "Preview list states" bar	Shown only with presenter=1 (S0-5)
Reset demo data as a single action	A dialog with Full demo and Empty schedule (S0-3)

10. Non-functional requirements

The reference laptop is a 2021 mid-range machine (4 cores, 8 GB RAM) running current Chrome, measured with the CPU throttled 4x for the stress tests.

10.1 Performance

ID	Requirement	Target
NF-P1	Load	Largest contentful paint 2.5 s or less at 10 Mbps. Initial JavaScript 300 kB gzip or less. Export libraries and the Excel parser load on first use.

ID	Requirement	Target
NF-P2	Schedule grid with 1,000 legs	Virtualised rows. First render 500 ms or less. Scrolling at 55 fps or better. Filter, search and sort respond in 100 ms or less.
NF-P3	Validation	A full run of R1 to R7 on 1,000 legs takes 100 ms or less. Revalidation after one edit takes 50 ms or less. Grouping by aircraft and date keeps the work near O(n log n).
NF-P4	Gantt	4 aircraft by 21 days drags at 60 fps. With 1,000 legs, lanes are virtualised and drag stays at 30 fps or better.
NF-P5	Import	90 legs parse in 3 s or less. 1,000 lines parse in 10 s or less with the interface still responsive, which means parsing runs in a Web Worker.
NF-P6	Export	12 legs in 3 s or less, 40 legs in 8 s or less, for each format
NF-P7	Reset	1 s or less

Stress mode: the URL parameter `legs=1000` repeats the 28-leg weekly pattern for 36 weeks, cut at 1,000 legs, with no planted exceptions. It exists for the performance tests and is not linked from the interface.

10.2 Accessibility

Target: WCAG 2.2 level AA on every screen used in the three storylines.

- NF-A1. Every function works by keyboard. Dragging a Gantt leg has a keyboard route through the leg drawer (S2-12). Focus is visible, drawers and dialogs trap focus, Esc closes them and focus returns to the control that opened them.
- NF-A2. Landmarks and headings are real. Tables use header cells with scope. The Gantt has an equivalent table view of the same data.
- NF-A3. Colour is never the only signal. Severity shows an icon and the word "Error" or "Warning". Availability cells carry the text Available, Maintenance or AOG. Text contrast is 4.5:1 and non-text contrast 3:1 in both light and dark themes.
- NF-A4. Toasts are announced politely (aria-live). Fix results say what changed, for example "PHC509 on 4L-PXB is now PHC509D".

- NF-A5. Interactive targets are at least 24 by 24 px. The layout holds at 200% zoom. prefers-reduced-motion turns the 2 s row highlight into a static outline.
- NF-A6. Icon-only buttons have accessible names. Word export uses real headings and tables.
- Check: axe-core on every screen with zero critical and zero serious findings, plus a manual keyboard pass of the three storylines.

10.3 Browser and device support

- NF-B1. Current and previous major versions of Chrome, Edge and Firefox on Windows, macOS and Linux, and Safari 17 or later on macOS. No Internet Explorer.
- NF-B2. Designed for 1440 by 900 and 1920 by 1080 (laptop and projector). Usable from 1280 px wide. Below 1024 px the app shows "Open on a larger screen".
- NF-B3. Body text is 14 px or larger so it reads on a projector. The app stays usable at 125% browser zoom.
- NF-B4. File downloads are tested in all four browsers, including file names with underscores and digits in Safari.

10.4 Reset-demo behaviour and data handling

- NF-R1. Three layers: the seed (immutable, in the bundle), the session state (memory plus `sessionStorage` key `aeroplanner.demo.v1`) and the interface state (filters, ticks, drawers).
- NF-R2. Edits survive a reload in the same tab. A new tab or a closed tab starts from the full demo. A stored version that does not match the app version is discarded silently.
- NF-R3. Reset to full demo replaces every store with the seed, clears filters, ticks, import state and the dismissed banner, restores the demo clock, opens Overview and announces the result in a live region. It takes 1 s or less.
- NF-R4. Reset to empty schedule keeps company, fleet, airports and maintenance blocks, and removes legs and permits. Importing the sample in this state yields the same 9 open issues as the full demo.
- NF-R5. Entry points: the top bar, the Conflicts empty state, `?reset=1` (full demo, then the parameter is removed) and `?start=empty`.
- NF-R6. There is no automatic reset on idle.
- NF-R7. No randomness or real clock in application logic. A lint rule forbids Math.random and Date.now outside the test helpers, so every run gives the same ids and results.

- NF-R8. An uploaded workbook never leaves the browser. The Import screen says so ("Your file stays in your browser"). No cookies and no third-party requests; fonts are self-hosted.
- NF-R9. Deployment: Next.js (App Router) with client components only, exported as a static site on Vercel. No API routes and no secrets. Production deploys from the main branch and every branch gets a preview URL. Responses carry `X-Robots-Tag: noindex` and robots.txt disallows everything. Vercel Deployment Protection (a password) is an option for prospect links.

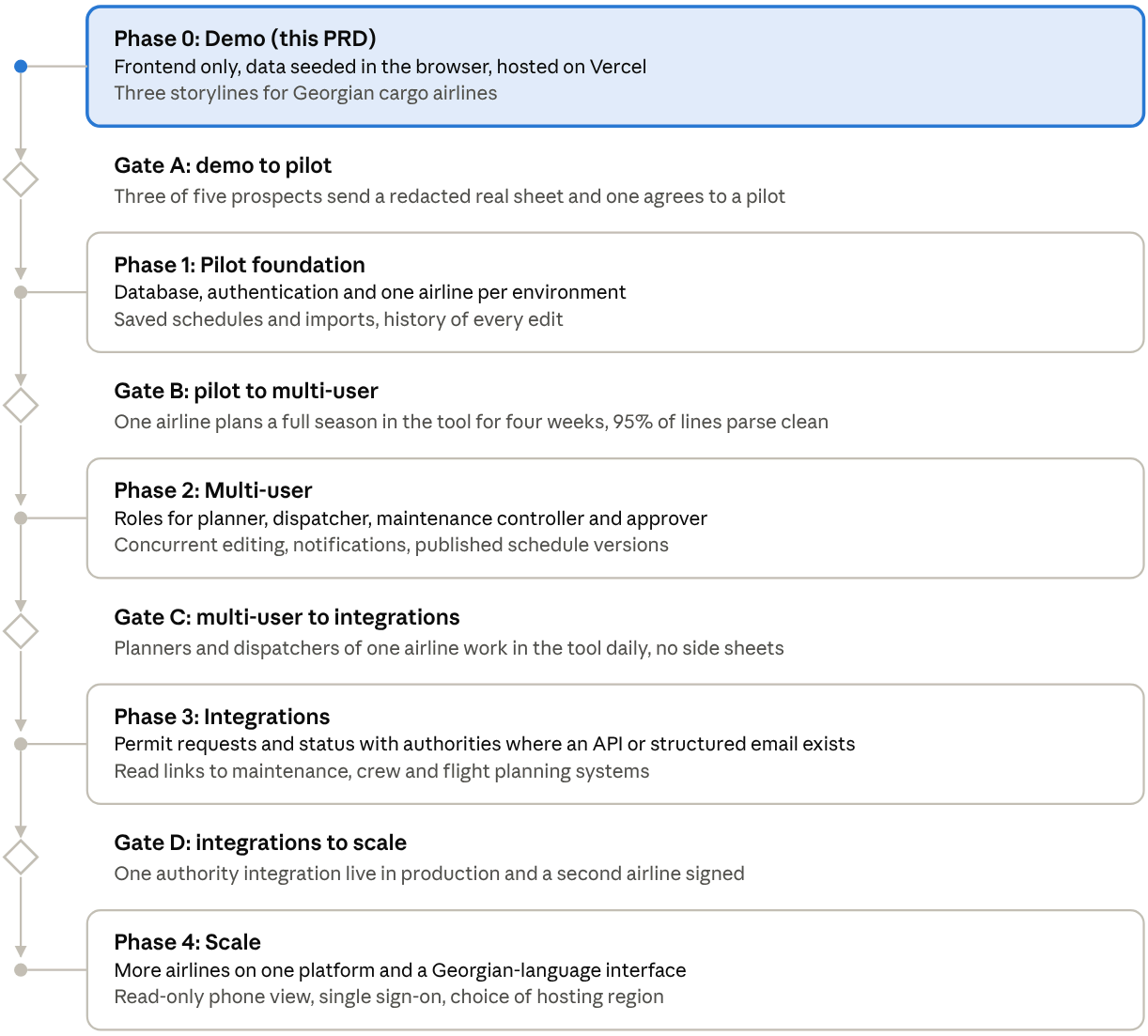
10.5 Quality gates in CI

A branch cannot merge unless these pass: type check, unit tests for every rule test (T-R1 to T-R7), parser tests P1 to P12, seed totals (9.8), Playwright scripts for storylines A, B and C, axe-core on all screens, the bundle budget (NF-P1) and a Lighthouse desktop performance score of 90 or more.

11. Roadmap beyond the demo and open questions

The demo proves the interface and the rules. The roadmap adds persistence, people and authorities in that order, and a phase starts only when the gate before it is met. Durations and team size are not set here, because they depend on what the first pilot airline needs.

Five phases from demo to scale, each gated by evidence from the one before



Gate criteria are proposals to confirm with the first pilot airline. Phases are in order, not to scale.

roadmap · 5 phases, 4 gates

Phase 1 replaces browser seeds with a database and real authentication, keeping the rule engine and parser from the demo unchanged. Phase 2 turns single-user edits into shared ones. Phase 3 connects to authorities and other systems only where a usable channel exists. Phase 4 adds tenants and languages.

Open questions

Each question lists the assumption this PRD uses until it is answered.

#	Question	Assumption in this PRD	Needed
1	What do the trailing "+" and "-" and the cell colours mean in the airlines' sheets?	"+" permits obtained, "-" pending, no marker means no record. Green, amber and red fill mean obtained, pending and cancelled.	Before build
2	What does "DAY n LT" count: ISO weekday of the local arrival date, day of the month, or a day of the operating pattern?	ISO weekday of the local arrival date	Before build
3	Do planners need both a scheduled (STD) and an estimated (ETD) time per leg?	One planned time per leg, and STD is read as ETD	Before pilot
4	Which prefix do real sheets use? The example string starts with CMS and the demo operator uses PHC.	A prefix that differs from the company designator is flagged with a suggestion	Before pitch
5	What is the minimum ground time, and does it differ by aircraft type or station?	60 minutes everywhere	Before pilot
6	Is a through-flight always one aircraft under one number, or can the aircraft change on the way?	R1 exempts a chain with an aircraft change	Before build
7	Is D the only parallel-flight suffix?	Only D	Before build
8	Which permit types exist beyond landing and overflight, and what number and validity formats does each authority use?	Landing and overflight only, free-text number, validity of the ETD date and the next day	Before pilot
9	Do prospects plan around daylight saving changes (Europe changes on 25 Oct 2026)?	Fixed offsets, no daylight saving	Before pilot
10	Must the interface be available in Georgian or Russian from day one?	English only	Before pitch

#	Question	Assumption in this PRD	Needed
11	Can two or three prospects share a redacted real sheet under NDA before the pitch, so the grammar in section 7 is checked against real data?	The layout in 7.1 is assumed from the design and one example string	Before pitch
12	Do prospects use an authority-approved flight packet template that the checklist must follow?	A generic layout under the company header	Before pilot
13	What hosting and data-residency rules apply to real schedules (EU, Georgia, on premise)?	Not relevant to the demo	Before phase 1
14	Is a password on the demo link acceptable to prospects?	Vercel Deployment Protection is optional	Before pitch